

Bailey Jannuzzi

PhD Student, Mathematical and Computational Biology
Department of Applied Science
William & Mary

Education

William & Mary | Applied Science Department Williamsburg, VA
PhD in Mathematical and Computational Biology Expected: 2031
Advisor: Geoffrey Zahn, PhD

Utah Valley University Orem, UT
Bachelor of Science in Bioinformatics | Magna Cum Laude Spring 2026
GPA: 3.81

Research Experience

Utah Valley University Orem, UT
Hjelmen Lab | Independent Student Researcher Spring 2024 - Present

Research Advisor: Carl E. Hjelmen, PhD
Investigating the diet of forensically relevant blow flies (Diptera: Calliphoridae) to determine their effectiveness as wildlife monitors.

- Constructing reproducible bioinformatic pipelines to identify environmental DNA from blow fly gut content samples.

Utah Valley University Orem, UT
Hjelmen Lab | Peer Research Collaborator Spring 2025 - Present

Research Advisor: Carl E. Hjelmen, PhD
Assisting with gathering data on blow fly (Diptera: Calliphoridae) diversity and distribution in Utah.

- Conducting field collections.
- Preparing collected specimens through pinning and labeling to be identified.

Presentations

*Presenting author

Poster Presentations

American Society for Microbiology, Intermountain Branch Meeting 2026
Evaluating Genomic Pipelines for Predicting Bacteriophage Host Range in Salmonella enteric, *Rogers N, Jannuzzi B, Johnston M, Lopez A, Brooks L

Entomological Society of America, Pacific Branch Meeting 2025
Oxford Nanopore Technology Sequencing and Identification of Fly Gut Content, *Jannuzzi B, Meeds A, Weidner LM, and Hjelmen CE

Entomological Society of America, Annual Meeting 2025
From City to Canyon: A Survey of Blow Fly Biodiversity Across Utah Ecoregions, *Beck H, Jannuzzi B, Weidner LM, and Hjelmen CE

Oral Presentations

North American Forensic Entomology Association, Annual Meeting 2026
From Gut to Database: Leveraging Blow Fly Dietary DNA for Forensic and Conservation Applications, *Jannuzzi B, Hornsby D, Meeds A, Weidner LM, and Hjelmen CE

North American Forensic Entomology Association, Annual Meeting	2026
<i>Characterizing Calliphoridae Diversity in the Mountain West: A Statewide Survey of Blow Fly Communities in Utah</i> , *Beck H, Jannuzzi B, Weidner LM, Hjelman CE.	
Entomological Society of America, Annual Meeting	2025
<i>Optimizing a Bioinformatic Pipeline for Profiling Blow Fly Diets</i> , *Jannuzzi B, Meeds A, Weidner LM, and Hjelman CE	
Entomological Society of America, Annual Meeting	2025
<i>What flies eat and excrete: Insights from puke, poop, and gut contents of blow flies for wildlife and forensic applications</i> , *Weidner LM, Jannuzzi B, Yeilee Perez Ortiz, Gabrielle Meza, Meeds A, and Hjelman CE	

Awards and Funding

Funding (Totaling \$3,719)

Undergraduate Research, Scholarly, and Creative Activity Dissemination Grant	2025
Amount Awarded: \$1,093	
Scholarly Activities Committee Grant	2025
Amount Awarded: \$1198	
Undergraduate Research, Scholarly, and Creative Activity Grant	2025
Amount Awarded: \$1428	

Awards and Honors

Outstanding Student Award for Biology — Utah Valley University, College of Science	Spring 2026
Undergraduate 10-Minute Presentation: SysEB, Behavior and Biodiversity 2nd Place Award	2025
Dean's List — Utah Valley University	Spring 2023 - 2026

Experience

Utah Valley University	Orem, UT
General Biology Teaching Assistant	Fall 2025
- Grading assignments and exams. - Hosting weekly study and review sessions. - Assisting with in-class activities. - Teaching vignette style lectures in-class.	
Utah Valley University	Orem, UT
Genetics Teaching Assistant	Spring 2025
- Supporting students in applying class concepts to solve practical problems.	

Membership

UVU Evolution and Bioinformatics Club President	Spring 2025 - 2026
UVU Evolution and Bioinformatics Club Secretary	Spring 2024 - 2025
American Society for Microbiology, Intermountain Branch	Current
North American Forensic Entomology Association	Current
Entomological Society of America	Current
Entomological Society of America, Pacific Branch	Spring 2025 - 2026

Skills

Computational Skills

R, Python, SQL, BASH Scripting/Linux, Bioinformatic Pipeline Development (Oxford Nanopore sequencing, eDNA metabarcoding)

Updated Jun 4, 2026

Laboratory Skills

DNA Extraction, PCR, gel electrophoresis, DNA purification, insect specimen preparation (pinning & labeling)